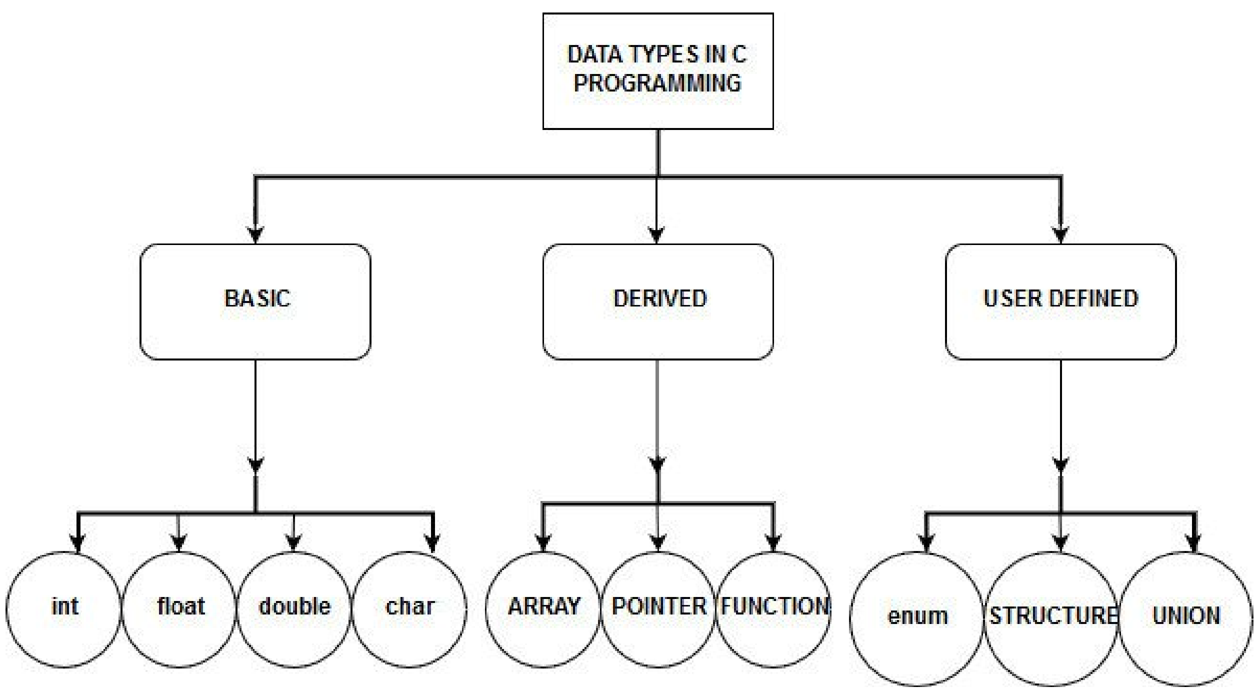
Data Types

C has a concept of 'data types' which are used to define a variable before its use. The definition of a variable will assign storage for the variable and define the type of data that will be held in the location. The value of a variable can be changed any time. The data types are the core building blocks in C.

C has the following basic built-in data types.

- int
- float
- char
- void



In general the overall data types are used in C/C++ are here:

- int (Integer)
- unsigned int
- long
- unsigned long
- long long int
- float
- double
- long double
- char (Character)
- unsigned char
- array
- pointer
- enum (Enumeration)
- struct (Structure)
- union

Data Types Range

Data Type	Memory (bytes)	Range	Format Specifier
short int	2	-32,768 to 32,767	%hd
unsigned short int	2	0 to 65,535	%hu
unsigned int	4	0 to 4,294,967,295	%u
int	4	-2,147,483,648 to 2,147,483,647	%d
long int	4	-2,147,483,648 to 2,147,483,647	%ld
unsigned long int	4	0 to 4,294,967,295	%lu
long long int	8	-(2^63) to (2^63)-1	%lld
unsigned long long int	8	0 to 18,446,744,073,709,551,615	%llu
signed char	1	-128 to 127	%c
unsigned char	1	0 to 255	%c
float	4		%f
double	8		%lf
long double	12		%Lf

int - data type

int is used to define integer numbers.

```
   {
int Count;
      Count = 5;
   }
```

float - data type

float is used to define floating point numbers.

```
{
   float Miles;
   Miles = 5.6;
}
```

The float type has a precision up to 7 digits. The double data type has a precision up to 15 digits.

char - data type

char defines characters.

```
{
   char Letter;
   Letter = 'x';
}
```

void type

void type means no value. This is usually used to specify the type of functions which returns nothing. Void can be as follows:
void main()

Array

An array is a collection of data items, all of the same type, accessed using a common name. Array works on integer and character value and can be declared as
type array_name [array_size]

Pointers

A **pointer** is a variable whose value is the address of another variable, i.e., direct address of the memory location. Like any variable or constant, you must declare a pointer before using it to store any variable address. The general form of a pointer variable declaration is as follows:
type *var_name

Enum (Enumeration)

Enumeration is a user defined datatype in C language. It is used to assign names to the integral constants which make a program easy to read and maintain. The enum is used for values that are not going to change (e.g., days of the week, colors in a rainbow, number of cards in a deck, etc.

```
enum day {
Saturday, Sunday, Monday, Tuesday, Thursday, Friday
};
```

Structure (struct)

Structure is a user-defined datatype in C language which allows us to combine data of different types together. Structure helps to construct a complex data type which is more meaningful. It is somewhat similar to an Array, but an array holds data of similar type only. But structure on the other hand, can store data of any type, which is practical more useful.

```
struct Student
{
char name[25];
int age;
char branch[10];
char gender;
};
```

Union

A union is a data type which has all values under it stored at a single address. It is a special data type available in **C** that allows storing different data types in the same memory location. You can define a **union** with many members, but only one member can contain a value at any given time. **Unions** provide an efficient way of using the same memory location for multiple-purpose

```
union car
{
char name[50];
int price;
}
```

Difference Between Structure and Union In C:	
C Structure	**C Union**
Structure allocates storage space for all its members separately.	Union allocates one common storage space for all its members. Union finds that which of its member needs high storage space over other members and allocates that much space
Structure occupies higher memory space.	Union occupies lower memory space over structure.
We can access all members of structure at a time.	We can access only one member of union at a time.

Structure example:
struct student
{
int mark;
char name[6];
double average;
};
	Union example:
	union student
	{
	int mark;
	char name[6];
	double average;
	};